JACK WESTBROOK

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EDUCATION

Imperial College London | MSc Pure Mathematics *2025-2026*

Awarded with Distinction; programme total: 79. Degree conferred 1 November 2026.

Dissertation: On Serre's Open Image Theorem (supervisor: Prof. George Boxer). Wrote an exposition of Serre's proof, with original worked examples in algebraic number theory and group theory.

University of Wisconsin-Madison | BA Mathematics *May 2025*

Overall GPA: 3.63/4.00 | Graduate-level mathematics coursework GPA: 3.91/4.00

RESEARCH AND ANALYTICAL EXPERIENCE

University of Michigan | Mathematics Research Assistant (REU) *Summer 2024*

- Developed a two-parameter family with varying Hilbert-Kunz multiplicities, combining density arguments and Segre-product constructions; coauthored a research manuscript.
- Contributed to a separate project on deformation of perfectoid purity in mixed characteristic; coauthored a submitted preprint (arXiv:2504.02966).

University of Wisconsin-Madison | Mathematics Research Assistant (REU) *Summer 2022*

- Used MATLAB to investigate Newton polygons for Lie algebras of vector fields through dimension 14, then developed combinatorial arguments addressing general dimensions.
- Presented findings at research venues and coauthored a manuscript in preparation.

SELECTED TECHNICAL AND MATHEMATICAL PROJECTS

Segre and Veronese Presentations | Macaulay2 software *2024-2025*

- Developed Macaulay2 packages to compute generators and defining relations for Segre products and Veronese subrings of graded rings; refined the implementation following mentor feedback.

Original Number Theory Problems | Independent project

- Designed nine original Diophantine and modular-arithmetic problems with complete solutions; one was selected for a UW-Madison Putnam Club meeting.

TECHNICAL SKILLS

Programming: Python, MATLAB, Macaulay2 (package development); familiarity with C++, SageMath, PARI/GP. Tools: Git, LaTeX.

SELECTED PUBLICATIONS AND LEADERSHIP

On Deformation of Perfectoid Purity in Gorenstein Domains (coauthor; submitted 2025; arXiv:2504.02966). Additional manuscripts on Hilbert-Kunz multiplicities and Newton polygons in preparation.

Directed Reading Program Mentor, Imperial College London (2025-2026); led undergraduate reading group on elliptic curves and Galois representations.